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RESEARCH ARTICLE

Exhaustivity and contrastivity in Hungarian focus constructions: A visual-world study

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Abstract
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1. Introduction

How languages express contrast and exhaustivity has major implications for how we understand the syntax–semantics interface. This study revisits the debate on the interpretation of Hungarian focus constructions using eye tracking and the visual-world paradigm. Specifically, we examine two theoretically important inquiries: how strongly Hungarian native speakers interpret phrases in the pre-verbal focus domain exhaustively and whether contrastive but non-exhaustive readings exist, as well as from a cognitive perspective, whether different focus types produce processing differences during language comprehension.

(1) *Anna* *meg-ette* *a* *banánt.*
Anna[NOM] VPX-eat-PST-3SG.DEF the banana-ACC
'Anna ate (up) the banana.'

(2) *Anna* *a* *BANÁNT* *ette* *meg.*
Anna[NOM] the banana-ACC eat-PST-3SG.DEF VPX
a) 'Anna ate (up) the BANANA.' [and nothing else]
b) 'Anna ate (up) the BANANA.' [and not something else]

Consider the canonical, neutral sentence in example 1, where the subject *Anna* is in the sentence-initial position, followed by the verb, *ette* 'ate' (together with the verbal prefix [VPX] (or verbal modifier [VM]) *meg-*)[1] and the object *banán* 'banana' (with the *-t* accusative ending); vs the pre-verbal focus construction in example 2, where the object is fronted to the pre-verbal position, leaving its (normally attached) verbal prefix behind. The sentence in example 2 is a typical example for a Hungarian focus construction, and it could also be rendered in English with an *it*-cleft, i.e., 'It is the BANANA[2] that Anna ate.'. The comparison of the Hungarian focus-constructions with English *it*-clefts has its own literature (cf. É. Kiss 1999; Wedgwood et al. 2006), but it is out of the scope of the present discussion.

Constituents in canonical sentences can be focused for various communicative purposes, such as emphasis, question answering, or contradiction. In 2 we are focusing the object, and as such we can observe two changes that mark focus: movement and prosody. Regarding the first, Hungarian has a special, grammaticalized syntactic position for the focused constituent directly in front of the verb – called the pre-verbal focus (PVF) position (Balogh 2007; Wedgwood 2006) or structural focus position (É.Kiss 1998) or simply

[1] Verbal prefixes in Hungarian can modify the meaning of the verb, or act as aspect markers. *Meg-* in this case has a perfective aspect meaning, indicating not only that the action has been concluded but that the event is complete, with the apple wholly eaten.
[2] Linguistically focused elements in the examples will be set in capitals for highlight throughout the paper.

just focus position – often associated with exhaustive/identificational semantics (Balogh 2013). Besides movement, the syntactically expressed focus is also accompanied by prosodic stress, and it has been suggested that in these focus marking strategies movement is primary, and prosody is secondary (Balogh 2013). Regarding intonation in PVF constructions, the focused constituent typically bears eradicating stress (cf. Gécseg 2020; Kornai & Kálmán 1988), meaning that the verb is destressed and the acoustic prominence of all other elements are leveled. The PVF construction is a common, everyday focus-marking strategy in Hungarian – a topic-prominent, discourse-configurational language where the configuration of constituents expresses information structural roles as opposed to grammatical ones (cf. É. Kiss 1995) – and various aspects of it has garnered significant attention in both theoretical and psycholinguistic research.

1.1. Background

One of the key issues around Hungarian focus is whether it is inherently exhaustive, or if the exhaustivity is a pragmatic inference. In other words, does the PVF construction itself entail that the focused element is the only one for which the predicate holds true, or can it be interpreted in a contrastive way, where the focused element stands in opposition to other alternatives without necessarily excluding them? For example, in 2 above, does this sentence implies that Anna ate only the banana *and nothing else* – an exhaustive reading, or could it mean that Anna ate the banana *and not something else* (or even, that she ate the banana but maybe *she also ate something else*) – a contrastive reading? We are interested in the 'default' interpretation of focus constructions such as these, and whether Hungarian speakers tend to understand them exhaustively or contrastively without additional context, as demonstrated by the two possible readings given in 2.

We should therefore try to define what we mean by *exhaustive focus* and *contrastive focus*. First, both types require the pre-verbal structure, but they differ in their semantic/pragmatic implications. In É.Kiss (1998)'s influential paper, both would fall under the term of *identificational focus*, which she defines as a focus that identifies the referent of the focused element from a set of alternatives. It takes scope, binds a variable, and interacts with quantification. É. Kiss treats contrastive focus as a subtype of identificational focus, where she describes it as [+exhaustive], and often [+contrastive]. However, we will argue that identificational focus does not always entail exhaustivity. According to Rooth (1992)'s theory on focus interpretation, exhaustivity is not inherent to focus but arises when focus interacts with specific operators like 'only'. For him, focus is a mechanism that introduces a set of alternatives (*alternative semantics*), and he considers contrast to be the pragmatic use of focus, where one alternative is highlighted against others in the set. In Vallduví & Vilkuna (1998), *kontrast* is an operator-like category that introduces a set of alternatives for the focused constituent. If an expression a is contrastive, a membership set $M = \{\ldots, a, \ldots\}$ is generated, serving as a quantificational domain. This allows alternative propositions $P(x)$ to be derived by substituting other members of M for a. Thus, *kontrast* provides the semantic mechanism by which focus evokes alternatives, enabling identificational or exhaustive readings. In Krifka (2008:259)'s work on Information Structure (IS), both exhaustive and contrastive focus is mentioned as subtypes of focus, and while exhaustive is interpreted as the only alternative that makes the proposition true narrowing the set of alternatives down to one (and thus enforcing exclusivity), contrastive focus is defined as a focus used for truly contrastive utterances and presupposes a competing alternative in the common ground, highlighting the contrast. Krifka's distinction aligns with his broader distinction between semantic (truth-conditional impact) and pragmatic (discourse management) uses of focus. Zimmermann (2008) also argued that contrastivity is not a semantic property of focus at all but rather a pragmatic one, and that it is best understood in terms of hearer expectation and discourse predictability, as a discourse-semantic device that signals unexpectedness relative to the hearer's assumptions, rather than a purely semantic operator over alternatives.

In short, exhaustive focus implies that the focused element is the sole entity for which the predicate holds true; i.e. that there are no other things Anna ate. On the other hand, contrastive focus would imply that the focused element stands in opposition to other possible alternatives, drawing attention to its uniqueness, unlikeliness, or contradicting nature given the context, e.g., 'Anna ate the banana [and not the apple].', but it does not necessarily exclude alternatives, i.e., there could be other things that Anna ate, e.g., 'Anna ate the banana [and/but also the apple].' Now consider the following:

(3) *Anna* ***csak*** *a* *BANÁNT* *ette* *meg.*
Anna[NOM] only the banana-ACC eat-PST-3SG.DEF VPX
'Anna ate (up) only the BANANA.'

(4) *Anna* ***többek között*** *a* *BANÁNT* *ette* *meg.*
Anna[NOM] others amongst the banana-ACC eat-PST-3SG.DEF VPX
'Anna ate (up), amongst other things, the BANANA.'

We can, in fact, force an exhaustive reading by adding the exhaustive focus marker *csak* 'only' to the sentence, as in example 3[3]. One intuitive reading of a simple PVF sentence like that in example 2 would be that 'Anna ate the banana [and nothing else].' In this sense, the sentence seems to be equal to that in 3, where the exhaustive interpretation is explicitly expressed with the operator *csak* 'only'. If we accept É.Kiss (1998)'s view, this seems to be redundant as the structural focus is already exhaustive. The question of differentiating the exhaustivity entailed by the pre-verbal focus structure vs the exhaustivity operator has been studied in detail by Balogh (2007).

Moreover, we can also cancel exhaustivity within the same clause with the help of certain phrases, such as *többek között* 'amongst others', as in example 4[4], by inserting a phrase with an inherently non-exhaustive, inclusive meaning (see also Balogh 2013; Wedgwood 2006:14). This sentence is not only perfectly grammatical, inclusive of alternatives, but also expresses a contrastive sense due the use of focus (there must be some specific discourse function for choosing to focus a constituent).

In Hungarian linguistic research, the role of semantic exhaustivity in focus constructions has been debated for more than forty years. The central question is whether exhaustivity arises from the syntax–semantics interface or from pragmatic and contextual factors. Earlier accounts within the generative framework assumed that the Hungarian pre-verbal focus structure necessarily entails semantic exhaustiveness, claiming that exhaustivity is an inherent property of the focus structure itself (É.Kiss 1998, 2007; Horvath 2008; Kenesei 2006; Szabolcsi 1981), which was more or less the received view up until the 2000s.

[3]The scope of *csak* 'only' only extends to the subsequent element, it can only refer to the object and not the following VP.

[4]Similarly, only the object is in the scope of *többek között* 'amongst others', it is not possible to interpret this sentence as if the verb is an alternative to other actions one can do to a banana.

More recently, this claim has since been challenged and shown to be untrue by both theoretical and psycholinguistic work – supported by experimental data – and it has been proposed that the exhaustive interpretation emerges from discourse-pragmatics, or conversational implicature (Balogh 2013; Káldi and Babarczy 2016; Onea 2009; Wedgwood et al. 2006). In introducing their argument against the claims of Szabolcsi (1981), É.Kiss (1998, 2007), and Horvath (2008) that 'an immediately pre-verbal focus in Hungarian is semantically exhaustive' or 'is interpreted exhaustively, i.e. as if it were in the scope of an exclusive like "only"', Onea & Beaver (2009:1) suggested in their first example[5] (analogous to our example 2) that the implicit reading of a sentence of this type of construction should be exhaustive, i.e. 'Anna ate the banana [and nothing else].' However, their analysis suggests that while such sentences often invite an exhaustive reading, this interpretation is not structurally encoded but contextually inferred.

Looking at example 2 however, we feel that the underlying implication is not necessarily that Anna ate *nothing* else, but it could be the case that she did not eat *something* else. Therefore, in our view, the interpretation might as well be contrastive.

To test this, and to address the theoretical debate, we designed an experiment within the visual world paradigm in light with recent experimental studies (Gerőcs et al. 2014; Káldi et al. 2021; Onea & Beaver 2009), where people are first presented with typical PVF sentences, and then given a binary choice that offers two contexts where both are correct and plausible: either A) a single-object image corresponding to an exhaustive interpretation, or B) a multi-object image implying a contrastive interpretation (see the image pair in Figure 1 accompanying examples 1-4). After reading a sentence, participants have to choose the image that best matches the sentence they just read, deciding which context/situation fits it better. Their choices should tell us about the preferred interpretation of the PVF constructions, whether it is more naturally interpreted as exhaustive or contrastive. Besides collecting behavioural data on the choices participants made, we also record their reading times and collect eye-tracking data during the picture verification phase, which hopefully can give us additional insights into the cognitive processes involved in interpreting the different types of PVF constructions, and whether different focus types produce different patterns of eye movements, such as more focused attention on the single-object image for exhaustive interpretations, or more distributed attention for contrastive interpretations.

[5] *Péter MARIT csókolta meg.* Peter Mari-ACC kiss-PST-3SG.DEF VPX 'Peter kissed MARY.'

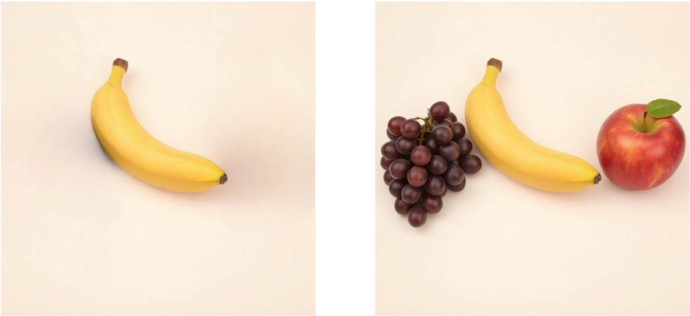

Figure 1. *Example stimuli for a sentence–picture verification task*

If the PVF construction is inherently exhaustive, we would expect participants to overwhelmingly choose the single-object image (type A) after reading the simple PVF sentences, and to show faster reaction times and more focused eye movements towards the single-object image. Conversely, if the PVF construction is more naturally contrastive, we would expect a more balanced distribution of choices between the single-object and multi-object images, and potentially different patterns of eye movements reflecting the consideration of multiple alternatives.

To sum up, while previous work has mainly tried to answer whether an immediately preverbal focus in Hungarian is semantically exhaustive or not, our question is: What is the default interpretation of a pre-verbal focus construction? Is it really inherently exhaustive, or could it be naturally contrastive?

2. Experiment

In this study, we are investigating focus in Hungarian, specifically how Hungarian speakers distinguish between exhaustive focus and contrastive focus – if they do, and how strongly they interpret various PVF constructions exhaustively. We designed a visual-world experiment with eye tracking to differentiate interpretations associated with the PVF structures. Specifically, we were curious about Hungarian speakers' dominant interpretation of the unmodified, simple PVF sentence (e.g., example 2), in comparison with the *only*–exhaustive (ex. 3) and the *amongst others*–contrastive conditions (ex. 4) For both of which we had some expectations: *only*–exhaustive sentences should be overwhelmingly interpreted as exhaustive, and *amongst others*–contrastive sentences should probably be interpreted as contrastive. The question was whether the unmodified PVF sentences would pattern with the exhaustive or the contrastive condition, or if they would show a more balanced distribution of interpretations.

2.1. Participants

We recruited 55 native speakers of Hungarian via the Prolific platform (www.prolific.com), where participants were pre-screened for Hungarian as their first language, and received a monetary compensation for their time (around £7.50/h). After rejecting 3 participants (due to overly long submission time, failing attention checks, or missing ET data), the final sample consisted of 52 individuals (32 female; mean age = 35). On average it took people just under 23 minutes to complete the experiment, although with a high individual variation (unusually long submission times were rejected).

2.2. Materials

Our stimuli consisted of 54 experimental sentences (18 items × 3 conditions) with 54 additional filler sentences in a randomized order (108 in total), but making sure that the same items never followed each other. Regarding lexical variation, we used verbs both with and without verbal prefixes (50-50%), and we included instances of both definite and indefinite conjugation (44.4-55.6%); also, both countable and uncountable nouns were used for object focus. We added some simple contextual fillers to the beginning of each sentence, such as *ami X-t illeti* 'regarding X' and time or place adverbials to balance sentence length

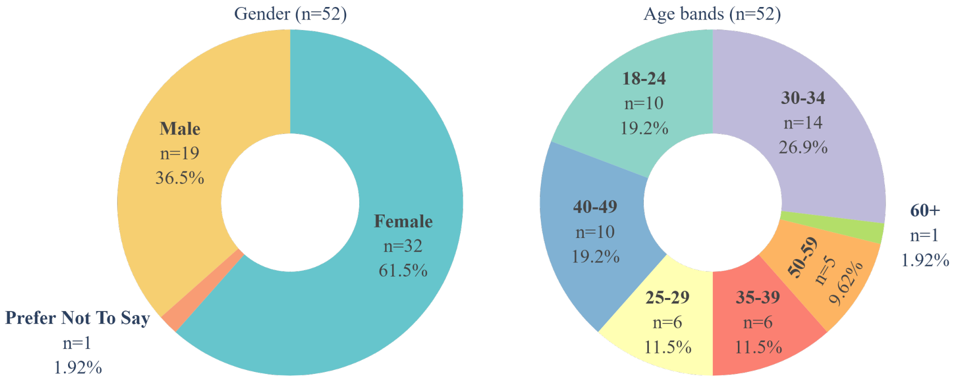

***Figure 2.** Demographic information of the participants in the experiment; gender (left) and age distributions (right)*

and make the targeted focus constructions appear not in the beginning of trials. When presenting the sentences on the screen, we divided them into regions, and articles were kept together with nouns, so all three conditions had the same number of regions (7), and the focused element was always in region 6 (r6). The stimuli were created by modifying a set of 18 base sentences – which were all simple transitive sentences – by modifying the base sentence with the addition of the focus marker *csak* 'only' for the exhaustive condition, and the phrase *többek között* 'amongst others' for the contrastive condition.

The 18 items were accompanied by 18 image pairs, these were created using DeepAI's image generator (https://deepai.org/) with prompts requesting simple 3D objects & scenarios in front of a light pastel background, and then modified manually to get matching pairs. The full stimuli list is available online, and in the Appendix we provide the full set of sentences and the image pairs (see Fig. 10).

To avoid overly long experimental sessions, we have split the experiment into three equal parts and each participant could complete one, two, or all three parts as separate events; meaning that we had three participant groups that could overlap, and each experiment consisted of 40 trials (36 experimental + 4 attention checks). This resulted in 72 valid submissions (25 for group one, 23 for group two, and 24 for group three), thus every stimuli was interacted with at least 23 times. The three parts were split in a way that for each group each item only appeared in one condition, and each session contained four attention check questions intermixed and evenly distributed across the trials.

The three conditions were: a) exhaustive focus with *csak* 'only' (e.g., 5a), b) unmodified, simple pre-verbal focus (PVF) (e.g., 5b), and c) contrastive focus with *többek között* 'amongst others' (e.g., 5c). Each item was paired with two images: A) one depicting a single-object interpretation – exhaustive reading – and B) one depicting a multiple-object interpretation – contrastive reading. See Figure 3 for an example of the image pairs accompanying the sentences in the experiment.

Figure 3. *Image pair for the sentence in examples 5a, 5b, & 5c*

(5) a. **exhaustive focus**

Ami az innivalót illeti, Eszter csak kávét ivott.
which[REL] the drink-ACC regard-3SG Eszter only coffee-ACC drink-PST-3SG
'Regarding drinks, Eszter only drank coffee.'

b. **unmodified pre-verbal focus**

Ami az innivalót illeti, Eszter ma kávét ivott.
which[REL] the drink-ACC regard-3SG Eszter today coffee-ACC drink-PST-3SG
'Regarding drinks, Eszter drank coffee today.'

c. **contrastive focus**

Ma reggel Eszter többek között kávét ivott.
today morning Eszter others amongst coffee-ACC drink-PST-3SG
'This morning, Eszter drank, amongst other things, coffee.'

2.3. Design

We built a sentence–picture verification (SPV) task, in which participants read a sentence and were asked to indicate one matching meaning-representation between two visual stimuli. First, participants saw a preview of the image pair to get familiar with the objects/context, on the left and right half of the screen (20% of the screen's width and 40% of the screen's height) on a canvas 50% larger than the images to mitigate the inaccuracies of the eye tracker. After pressing the spacebar they were presented with the sentence word by word (region by region) in the middle of the screen in a self-paced reading (SPR) manner, in which participants had to use the spacebar to advance throught the sentences. We recorded reading times throughout the reading. After reading through the sentence, the image pair re-appeared and participants were asked to select the image that best matched the meaning of the sentence they just read. The choice had to be made with a mouse-click anywhere on the corresponding canvas (left or right), and reaction times (RT) and eye movements were recorded during this picture verification phase. The image types were counterbalanced throughout the trials. The segments were separated by 1000 ms pauses, and after the choice, the next trial started after a brief (1000 ms) pause. The experiment was designed to be completed in around 25-30 minutes, and participants had to complete it in one setting. See Figure 4 for a visual representation of the design of the experiment.

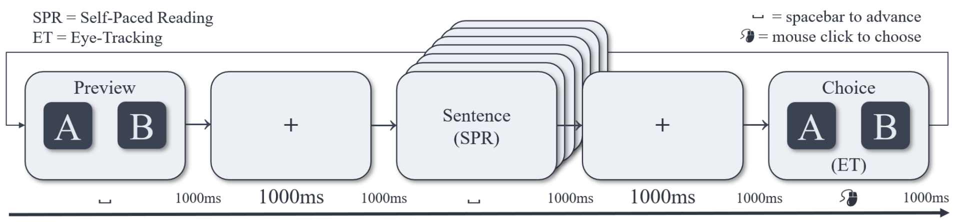

Figure 4. *Design*

2.4. Procedure

The experiment was created, hosted on, and presented (in Hungarian) to participants on PCIbex Farm (Schwarz & Zehr 2021), the code and resources are available online for inspection. Participants completed the experiment on their own devices (PC only, with a webcam and physical keyboard & mouse) and on their own time, but with Prolific helping us control for too fast or too slow submissions resulting in rejection or timeout.

First, potential participants were instructed to be in a well-lit environment, sit at an appropriate distance from their screen and webcam, and minimize head movements, emphasizing the need for a bright environment, and that they can only start the experiment from a Chrome browser. Before starting the experiment, participants then had to calibrate the eye tracker, which involves looking at specific points on the screen (e.g., corners, center) while the system records their eye positions to create a mapping between gaze coordinates and screen coordinates. Eye-tracking data were collected using the built-in WebGazer.js library of PCIbex, which uses the participant's webcam to track eye movements, and we used the default calibration procedure provided (a 9-point calibration grid). Participants had to reach and maintain at least 60% accuracy in the calibration phase to proceed with the experiment, and had three attempts to achieve this accuracy, ensuring that the eye-tracking data collected during the picture verification phase would be reliable. The tracker also checked their gaze again and again before every trial, and if the accuracy dropped below 60%, participants were prompted to recalibrate before proceeding to the next trial. People who could not pass the calibration for various reasons (e.g., computer setup, bad lighing/environment, glare on glasses, or any other technical issue) were not able to start the experiment. After calibration, participants were intstructed on the exact task and how to complete the trials, followed by a practice session consisting of three trials, and finally the live experiment of 40 trials (including 4 attention checks).

We have to mention that a relatively large amount of people (around a third of all who tried) could not pass calibration, and thus were not able to start the experiment. This is a persistent issue with webcam-based online eye tracking studies, as it can be affected by various factors such as lighting conditions, webcam quality, and even individual differences in facial features. Depending on these factors, participants' experiences were vastly different, some people breezed through trials and had a smooth experience finishing under 20 minutes, others struggled and had to recalibrate from time to time, spending over 30-40 minutes in total. We urged participants to reach out if they had issues, but sadly we couldn't assist everyone remotely. A small number of participants started but couldn't finish the experiment due to technical reasons, they were also compensated for their time, but their data was not included in the analysis.

2.5. Analysis

2.5.1. Behavioral data?

2.5.2. Eye-tracking data processing

Eye-tracking data recorded on PCIbex comes in a relatively straightforward, clean format not of gaze coordinates but in classified samples, where each sample is already defined as a gaze at a canvas or not, in our case: left canvas, right canvas, or neither. Samples are recorded between every 30-70 milliseconds, depending on the users' computer & browser setup, internet speed, and presumably the general load that the PCIbex server was under at the time.

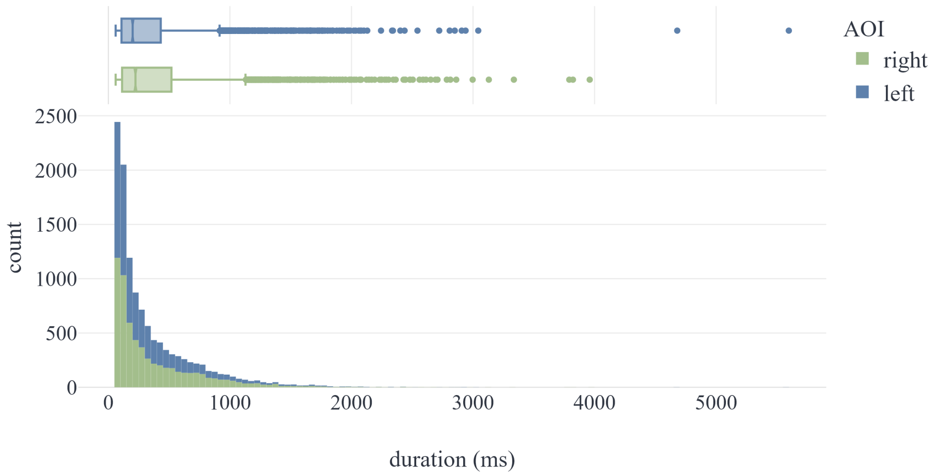

Figure 5. *Distribution of episode durations after cleaning and before winsorization*

The following steps were taken to clean and classify the eye-tracking data in order to obtain a certain set of metrics that could be used for analysis: First, raw eye tracking data files were loaded per participant, and **samples** were classified into AOIs at each timestamp as left (left canvas=1, right canvas=0), right (right canvas=1, left canvas=0), or neutral otherwise. Then, same-side consecutive samples were collapsed into contiguous gaze **episodes** within participant and trial. Episode duration was computed from timestamp-derived sample durations where episodes shorter than a threshold of 60 ms were removed, and neutral-gap bridging was enabled (with a max neutral bridge of 50 ms). In other words, very short gaze episodes and rapid saccades were removed cleaning noise & jitter, as they are unlikely to reflect meaningful cognitive efforts and more likely to be artifacts of the eye-tracking system. Then, neutral episodes were dropped and excluded from further analysis. Episodes thus intend to represent a moment of sustained gaze within an AOI, ignoring brief out-of-canvas samples that result from technical inaccuracies and are not deliberate (or uncounscious) eye movements. For quality control, episode-duration distributions were inspected via summary statistics, and episodes longer than the mean +2 SD ($\mu \pm 2\sigma$) were winsorized by replacing their duration with the global mean (i.e., with rows retained). This procedure preserved the full number of episodes while limiting the influence of extreme outliers; 4.89% of all episodes were substituted. See Figure 5 for the distribution of episode durations. This cleaning process was devised after established eye-tracking data classification protocols, such as the Identification by Dispersion-Threshold (I-DT) algorithm, which itself is ideal for low- to mid-frequency eye trackers and static stimuli.

Finally, trial-level eye-tracking (ET) features were computed, including left/right/total dwell time, dwell proportions, dominant side, fixation counts, revisits, transitions, and first-fixation side/duration. To briefly define the most important metrics that were used in the end: **dwell** is a time value, the sum of the duration of all gaze episodes within a trial (either left, right, or total), representing the total amount of time a participant has spent looking at the AOI(s) during the choice; **fixation** on the other hand is a numeric value, representing how many times a participant looked at (or switched between) the AOIs during a choice. Missing trial-level metrics were mean-imputed in 5.02% of the data points, which were mostly due to missing episode-level data (e.g., no episodes detected for a trial, or no episodes remained after cleaning).

We acknowledge that several processing decisions may limit inference. Gaze data was essentially collapsed to binary AOIs (left/right), while off-target samples were treated as neutral and excluded. Episodes shorter than 50 ms were removed, potentially excluding very brief but real events. Long-duration episodes were winsorized by replacing values above a global mean+2SD threshold with the global mean, preserving sample size but reducing variance and potentially biasing duration-based measures. Also, missing merged ET trial metrics were mean-imputed for selected variables, which further shrinks the effective sample size. We therefore interpret absolute magnitudes cautiously and focus on relative contrasts between conditions.

The eye-tracking data cleaning and classification pipeline was implemented in Python, and can be inspected online.

2.5.3. Statistical analysis

We analyzed two eye-tracking dependent measures: *total dwell time* (`total_dwell`) and *total fixation count* (`total_fixation`). For both measures, analyses were conducted in R using linear mixed-effects models (LMM) (packages `lme4`, `lmerTest`, `car`, and `emmeans`). The fixed-effect predictors were focus `Condition` (three levels: exhaustive, unmodified, contrastive) and image `Choice` (two levels: single-object, multi-object). To stabilize variance and accommodate zero values, we applied a log transform using $\log(1 + x)$ to the raw dependent measure prior to modeling. For each dependent measure, we fit the following model with `Participants` and `Items` as random intercepts: `y' ~Condition * Choice + (1 | Participant) + (1 | Item)` where $y' = \log(1 + y)$ is the transformed dependent variable. The model were fit using maximum likelihood estimation (i.e., `REML = FALSE`) to enable likelihood-ratio model comparisons. To follow up on significant effects in the model, we computed estimated marginal means and conducted Tukey-adjusted pairwise comparisons using `emmeans`. Specifically, we tested (i) pairwise differences between conditions within each type (Condition | Choice), (ii) the pairwise differences between types within each condition (Choice | Condition), and (iii) the full interplay of both (Condition $*$ Choice).

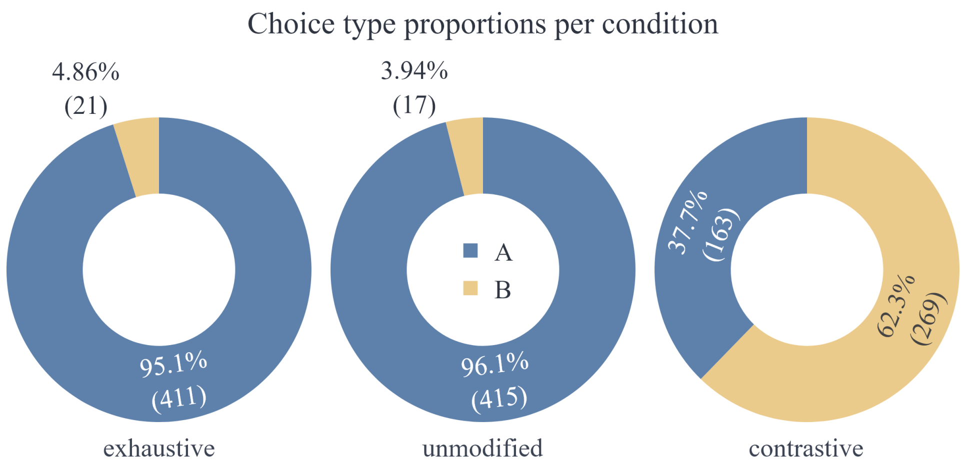

Figure 6. *Distribution of chosen visual stimuli (Image type A vs B) per condition*

3. Results

3.1. Behavioral results

Table 1. *Type of chosen image per condition.*

Condition	Image A: % (N)	Image B: % (N)
Exhaustive	95.1 (411)	4.86 (21)
Unmodified	96.1 (415)	3.94 (17)
Contrastive	37.7 (163)	62.3 (269)

The behavioural results showed that participants overwhelmingly interpreted the unmodified pre-verbal focus (PVF) sentences like exhaustive *only*-sentences, choosing single-object type A images (see Table 1, Figure 6). However, in the contrastive condition (*among others*-sentences) people preferred the multi-object type B images, exhibiting a tendency toward non-exhaustive readings, but with almost 37.7% of the choices – somewhat unexpectedly – still being cast on the exhaustive interpretation type A images. This suggests that even with the presence of a phrase that licenses an inclusive, contrastive reading, the exhaustive effect is still quite dominant. This is clear evidence that the PVF construction is not necessarily interpreted exclusively exhaustively, a non-exhaustive interpretation is still available, a contrastive reading can be be teased out easily, and also that the presence of a contrastive licensing phrase does not completely eliminate the exhaustive bias.

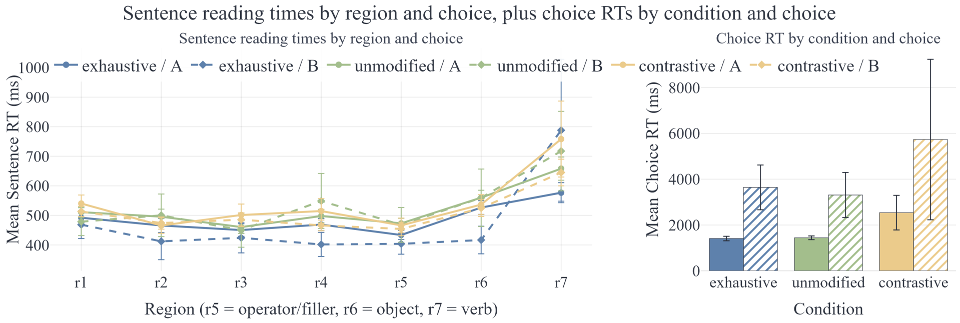

Figure 7. *Mean reading times of sentences by regions and reaction times of choices.*

3.2. Reaction Times

We analyzed the reading and reaction times (RT) for both sentence reading and the picture verification choices, see Figure 7 for an illustrative overview.

3.2.1. Sentence Reading Times

In the examination of the factors to see what is contributing to significant effect, the Type III Wald chi-square tests showed a significant main effect of Condition ($\chi^2(2) = 18.66, p <$ .001), but no main effect of Choice ($\chi^2(1) = 0.06$, $p = .806$) and no Condition × Choice interaction ($\chi^2(2) = 2.10$, $p = .351$) (see Table 2). This suggests that the different focus conditions had a significant impact on the overall reading times of the sentences, but the type of choice participants made (single-object vs multi-object) did not significantly affect their reading times. In other words, while the focus type influenced how long participants took to read the sentences, it did not seem to influence their decision-making process in terms of which image they chose, nor did the choice they made influence their reading times in a way that interacted with the condition.

Table 2. *Type III Wald chi-square tests for whole-sentence reading times.*

Effect	χ^2	df	p
Condition	18.66	2	< .001
Choice	0.06	1	.806
Condition × Choice	2.10	2	.351

Post-hoc pairwise comparisons revealed that the significant effect of Condition was driven by significantly faster reading times in the exhaustive condition compared to the contrastive condition (p=0.0004). This suggests that participants read the sentences with the exhaustive focus marker *csak* ‘only’ faster than the sentences with the contrastive phrase *többek között* ‘amongst others’, which indicates that the exhaustive sentences were easier to process or more expected, while the contrastive sentences may have required more cognitive effort or were less predictable, leading to longer reading times. There was no significant difference in whole sentence reading times between the exhaustive and unmodified conditions (p=0.99), nor between the unmodified and contrastive conditions (p=0.08), suggesting that the presence of the focus marker *csak* 'only' did not significantly speed up reading compared to the unmodified PVF sentences, and the unmodified sentences did not differ significantly from the contrastive sentences in terms of reading times.

Table 3. *Tukey-adjusted pairwise comparisons for whole-**sentence** reading times across Condition × Choice combinations.*

Contrast	Estimate	SE	df	*t*	*p*
Exhaustive A – Unmodified A	-0.05883	0.0140	1212	-4.196	.0004
Exhaustive A – Contrastive A	-0.04833	0.0200	1224	-2.422	.1494
Exhaustive A – Exhaustive B	0.01194	0.0486	1225	0.246	.9999
Exhaustive A – Unmodified B	0.05022	0.0529	1222	0.948	.9337
Exhaustive A – Contrastive B	-0.00913	0.0163	1218	-0.560	.9935
Unmodified A – Contrastive A	0.01050	0.0199	1223	0.529	.9950
Unmodified A – Exhaustive B	0.07077	0.0484	1224	1.463	.6882
Unmodified A – Unmodified B	0.10905	0.0530	1223	2.056	.3115
Unmodified A – Contrastive B	0.04970	0.0163	1218	3.046	.0286
Contrastive A – Exhaustive B	0.06026	0.0499	1223	1.208	.8330
Contrastive A – Unmodified B	0.09855	0.0548	1224	1.797	.4681
Contrastive A – Contrastive B	0.03920	0.0229	1233	1.712	.5237
Exhaustive B – Unmodified B	0.03828	0.0685	1220	0.559	.9935
Exhaustive B – Contrastive B	-0.02106	0.0492	1225	-0.428	.9982
Unmodified B – Contrastive B	-0.05935	0.0534	1221	-1.112	.8764

3.2.2. Regions Reading Times

Table 4. *Type III Wald chi-square tests for region 5 reading times.*

Effect	χ^2	df	*p*
Condition	18.66	2	< .001
Choice	0.25	1	.615
Condition × Choice	1.72	2	.423

Region-level reading times were analyzed at three critical regions: r5 (containing the modifying phrases *only*, a filler word, or the second half of *among others*), r6 (the focused object), and r7 (the sentence-final verbs).

In r5, there was a significant effect of Condition ($p < .001$), with r5 being read significantly faster in the exhaustive condition than in the unmodified condition ($p < .001$), showing that *csak* 'only' was processed more quickly than the corresponding region in the unmodified sentences, but not significantly quicker than the contrastive modifier *többek között* ‘amongst others’ (see Table 4).

Table 5. *Tukey-adjusted pairwise comparisons for* ***region 5*** *reading times across Condition × Choice combinations.*

Contrast	Estimate	SE	df	t	p
Exhaustive A – Unmodified A	-0.06953	0.0162	1213	-4.303	.0003
Exhaustive A – Contrastive A	-0.02961	0.0230	1232	-1.289	.7910
Exhaustive A – Exhaustive B	-0.02806	0.0559	1234	-0.502	.9961
Exhaustive A – Unmodified B	0.00644	0.0609	1230	0.106	1.0000
Exhaustive A – Contrastive B	0.00184	0.0188	1222	0.098	1.0000
Unmodified A – Contrastive A	0.03992	0.0229	1231	1.746	.5013
Unmodified A – Exhaustive B	0.04148	0.0557	1233	0.745	.9762
Unmodified A – Unmodified B	0.07598	0.0611	1230	1.244	.8148
Unmodified A – Contrastive B	0.07137	0.0188	1223	3.797	.0021
Contrastive A – Exhaustive B	0.00155	0.0574	1231	0.027	1.0000
Contrastive A – Unmodified B	0.03605	0.0631	1232	0.571	.9929
Contrastive A – Contrastive B	0.03145	0.0263	1246	1.195	.8392
Exhaustive B – Unmodified B	0.03450	0.0788	1225	0.438	.9980
Exhaustive B – Contrastive B	0.02989	0.0567	1234	0.528	.9951
Unmodified B – Contrastive B	-0.00461	0.0614	1228	-0.075	1.0000

For r6, which is the focused object, there was a significant effect of Condition ($p = .005$) and Choice ($p = .034$), but no significant interaction (see Table 6). This suggests that the reading times for the focused object were influenced by both the focus condition and the type of choice participants made, but these effects did not interact with each other. In other words, while both the focus condition and the choice type had an impact on how long participants took to read the focused object, the way these two factors influenced reading times did not depend on each other. The significant effect of Condition indicates that the different focus conditions led to different reading times for the focused object, while the significant effect of Choice suggests that whether participants ultimately chose the single-object or multi-object image also influenced their reading times for that region.

Table 6. *Type III Wald chi-square tests for region 6 reading times.*

Effect	χ^2	df	p
Condition	10.52	2	.005
Choice	4.52	1	.034
Condition × Choice	2.37	2	.306

Pairwise comparisons revealed that the significant effect of Condition in r6 was driven by significantly faster reading times in the exhaustive condition compared to the unmodified condition ($p = .002$), but no significant differences between the exhaustive and contrastive conditions was found ($p = .99$) which might be a residual effect of the scope of marker *only*.

Table 7. *Tukey-adjusted pairwise comparisons for **region 6** reading times across Condition × Choice combinations.*

Contrast	Estimate	SE	df	t	p
Exhaustive A – Unmodified A	-0.0620	0.0195	1212	-3.187	.0184
Exhaustive A – Contrastive A	-0.0170	0.0277	1230	-0.614	.9900
Exhaustive A – Exhaustive B	0.1429	0.0674	1232	2.120	.2775
Exhaustive A – Unmodified B	0.0973	0.0734	1228	1.326	.7709
Exhaustive A – Contrastive B	0.0415	0.0226	1221	1.834	.4438
Unmodified A – Contrastive A	0.0450	0.0275	1229	1.635	.5752
Unmodified A – Exhaustive B	0.2049	0.0671	1231	3.055	.0278
Unmodified A – Unmodified B	0.1594	0.0736	1228	2.167	.2543
Unmodified A – Contrastive B	0.1035	0.0226	1222	4.574	.0001
Contrastive A – Exhaustive B	0.1598	0.0692	1229	2.311	.1902
Contrastive A – Unmodified B	0.1143	0.0760	1230	1.503	.6622
Contrastive A – Contrastive B	0.0585	0.0317	1244	1.846	.4365
Exhaustive B – Unmodified B	-0.0455	0.0950	1224	-0.479	.9969
Exhaustive B – Contrastive B	-0.1013	0.0683	1232	-1.485	.6742
Unmodified B – Contrastive B	-0.0558	0.0740	1227	-0.754	.9749

Finishing the sentence with the unstressed verb in r7, there was a significant effect of Condition ($p < .001$), but no significant effect of Choice ($p = .453$) and no significant interaction (see Table 8).

Table 8. *Type III Wald chi-square tests for region 7 reading times.*

Effect	χ^2	df	p
Condition	25.48	2	< .001
Choice	0.56	1	.453
Condition × Choice	2.32	2	.314

Post-hoc pairwise comparisons revealed that the significant effect of Condition in r7 was driven by significantly faster reading times in the exhaustive condition compared to the unmodified condition ($p < .001$) and the contrastive condition ($p = .017$), but no significant difference between the unmodified and contrastive conditions was found ($p = .60$). This suggests that sentences were read faster in the final region only when readers tried to select A during reading the exhaustive condition as the last region typically reflects all preceding words' effects on comprehension.

Table 9. *Tukey-adjusted pairwise comparisons for* ***region 7*** *reading times across Condition × Choice combinations.*

Contrast	Estimate	SE	df	*t*	*p*
Exhaustive A – Unmodified A	-0.11708	0.0245	1213	-4.779	< .0001
Exhaustive A – Contrastive A	-0.11131	0.0348	1233	-3.198	.0177
Exhaustive A – Exhaustive B	-0.06344	0.0848	1234	-0.748	.9757
Exhaustive A – Unmodified B	-0.01551	0.0924	1230	-0.168	1.0000
Exhaustive A – Contrastive B	-0.04531	0.0285	1223	-1.591	.6047
Unmodified A – Contrastive A	0.00577	0.0346	1232	0.167	1.0000
Unmodified A – Exhaustive B	0.05364	0.0844	1233	0.636	.9883
Unmodified A – Unmodified B	0.10157	0.0925	1230	1.098	.8825
Unmodified A – Contrastive B	0.07177	0.0285	1223	2.519	.1192
Contrastive A – Exhaustive B	0.04787	0.0870	1232	0.550	.9940
Contrastive A – Unmodified B	0.09580	0.0957	1232	1.001	.9176
Contrastive A – Contrastive B	0.06599	0.0399	1248	1.656	.5616
Exhaustive B – Unmodified B	0.04793	0.1200	1225	0.401	.9987
Exhaustive B – Contrastive B	0.01812	0.0859	1235	0.211	.9999
Unmodified B – Contrastive B	-0.02980	0.0931	1229	-0.320	.9996

3.2.3. Decision Reaction Times

When it came to decisions, the Type III Wald chi-square tests revealed significant main effects of Condition (p = .018) and Choice (p = .001), as well as a significant interaction between Condition and Choice (p = .029) (see Table 10). This suggests that the task itself has influence on the decision time, unlike while previously, as expected, the choice did not show significant influence on reading times.

Table 10. *Type III Wald chi-square tests for choice reaction times.*

Effect	χ^2	df	*p*
Condition	8.04	2	.018
Choice	10.66	1	.001
Condition × Choice	7.06	2	.029

Pairwise comparisons revealed that the significant interaction between Condition and Choice was driven by significantly faster decision times in the exhaustive condition compared to the contrastive condition when participants chose A (p = .060), but no significant differences between exhaustive and unmodified conditions were found in this case (p = .691). During reading, sentences with *only* were significantly different from unmodified in most words. This can be interpreted as *only* sentences require an explicit exhaustive focus denotation, while the unmodified sentences did not specify the focus type. When it is time to make the decision, the difference is not significant, suggesting that it is easy to decide the reading of these two conditions.

Within conditions, there was a significant difference between choosing A and B in the exhaustive condition ($p = .015$), but not between others (both ps > .46). This means that only in the exhaustive condition the difference is obvious, but in unmodified and contrastive ones, A-B readings might both be possible!

No significant differences were found when participants chose B (all ps > .55). This suggests that the presence of the focus marker *csak* 'only' facilitated faster decision times when participants ultimately selected the single-object image (A), while the contrastive modifier *többek között* 'amongst others' did not provide the same facilitation, leading to slower decision times in that condition. However, when participants chose the multi-object image (B), there were no significant differences in decision times across conditions, indicating that the type of focus marking did not influence decision times for those choices.

Table 11. *Tukey-adjusted pairwise comparisons for* ***choice reaction times*** *across Condition × Choice combinations.*

Contrast	Estimate	SE	df	*t*	*p*
Exhaustive A – Unmodified A	-0.0489	0.0335	1213	-1.458	.6910
Exhaustive A – Contrastive A	-0.1325	0.0476	1241	-2.786	.0604
Exhaustive A – Exhaustive B	-0.3774	0.1160	1242	-3.256	.0147
Exhaustive A – Unmodified B	-0.2769	0.1260	1236	-2.192	.2422
Exhaustive A – Contrastive B	-0.1822	0.0390	1227	-4.677	< .0001
Unmodified A – Contrastive A	-0.0836	0.0474	1240	-1.765	.4889
Unmodified A – Exhaustive B	-0.3284	0.1150	1240	-2.847	.0510
Unmodified A – Unmodified B	-0.2280	0.1270	1237	-1.801	.4651
Unmodified A – Contrastive B	-0.1333	0.0390	1228	-3.421	.0084
Contrastive A – Exhaustive B	-0.2449	0.1190	1238	-2.058	.3103
Contrastive A – Unmodified B	-0.1444	0.1310	1238	-1.104	.8800
Contrastive A – Contrastive B	-0.0497	0.0544	1262	-0.915	.9428
Exhaustive B – Unmodified B	0.1005	0.1640	1228	0.614	.9900
Exhaustive B – Contrastive B	0.1951	0.1170	1242	1.662	.5570
Unmodified B – Contrastive B	0.0947	0.1270	1235	0.744	.9764

3.3. Eye-tracking results

To further examine how the focus structures were associated with exhaustive and contrastive readings, and how the conditions and choices influenced cognitive processing, we fitted linear mixed-effects models to two eye-tracking metrics: *total dwell time*, and *total fixation count* with fixed effects of Condition (exclusive–*only*, unmodified, and contrastive–*among others*), Choice (single- vs multi-object type images), and their interaction, with random intercepts for Participant and Item. Dependent variables were log-transformed using `log1p`. Type-III Wald χ^2 tests and Tukey-adjusted pairwise comparisons from `emmeans` were used for inference.

For *total dwell time* – the total duration of all gaze episodes on the AOIs during the choice – we found significant main effects of Condition: $\chi^2(2) = 12.531$, $p = 0.00190$

Table 12. *Type-III Wald χ^2 tests from mixed models.*

Metric	Effect	χ^2	df	p
total_dwell	condition	12.531	2	.0019 **
	choice	11.845	1	.0006 ***
	condition × choice	12.292	2	.0021 **
total_fixation	condition	12.326	2	.0021 **
	choice	3.346	1	.0674 .
	condition × choice	6.764	2	.0340 *

and Choice: $\chi^2(1) = 11.845$, $p = 0.000578$, as well as a significant interaction of Condition×Choice: $\chi^2(2) = 12.292$, $p = 0.00214$ (Table 12). Post-hoc (Tukey-adjusted) comparisons indicated that in the exhaustive condition, participants have spent significantly less time looking at the AOIs when they chose the A-type, single-object image (A–B differed: $\hat{\beta} = -0.4443$, SE $= 0.1290$, $t(1259) = -3.432$, $p = 0.0006$) signifying that the expected choice was not only easy but also a fast one, which makes sense for this control-like condition (see Figure 13). Follow-up pairwise comparisons also showed that participants who ultimately commited to the A-type, single-object image have spent significantly more time looking at the AOIs in the contrastive condition than in the exhaustive and unmodified focus conditions (see Table 14). This suggests that the contrastive condition elicited more cognitive effort and uncertainty when participants chose the single-object images, meaning that they have pondered longer on the decision, presumably finding it a bit more difficult. In contrast, for those who ended up choosing the B-type, multi-object image, there was no significant difference between the conditions, except a marginal trend for less gaze times in the contrastive condition when pitted against the exhaustive sentences.

For *total fixation count*, the trends were similar but much weaker. The Condition and the Condition×Choice interaction were significant, while the main effect of Choice was not (Table 12). Pairwise contrasts for choice within condition were not significant, and condition differences were again restricted to type A, where the choices after reading contrastive sentences elicited significantly more fixations than the exhaustive and unmodified conditions, hinting on participants being more puzzled and taking more looks to decide, in line with the dwell times (Tables 13–14).

Together with the behavioural choice pattern, these results suggest a robust exhaustive bias for PVF when no additional contrastive licensing is present, and increased processing cost when an exhaustive (type A) choice is made in a contrastive context.

4. Discussion

I ASKED AI TO HELP ME WITH THE DISCUSSION, ALL THE TEXT BELOW IS AI-GENERATED, I WAS CURIOUS WHAT WOULD IT SAY ABOUT OUR STUDY SO FAR WHEN I PROMPT FOR A DISCUSSION.

This study set out to address two related questions in the debate on Hungarian preverbal focus (PVF): (i) whether PVF is interpreted exhaustively by default, and (ii) whether

Table 13. *Tukey-adjusted pairwise contrasts for* ***choice within each condition*** *(A - B). SE = standard error; df = degrees of freedom.*

Metric	Condition	Estimate	SE	df	*t*	*p*
total_dwell	exhaustive	-0.4443	0.1290	1259	-3.432	.0006 ***
	unmodified	-0.2390	0.1410	1251	-1.689	.0914
	contrastive	0.0337	0.0605	1283	0.557	.5778
total_fixation	exhaustive	-0.2016	0.1110	1259	-1.824	.0684
	unmodified	-0.1254	0.1210	1252	-1.038	.2995
	contrastive	0.0848	0.0517	1284	1.640	.1012

Table 14. *Tukey-adjusted pairwise contrasts for* ***condition within each choice****. SE = standard error; df = degrees of freedom.*

Metric	Contrast	Estimate	SE	df	*t*	*p*
choice A						
total_dwell	exhaustive - unmodified	-0.0474	0.0376	1213	-1.258	.4193
	exhaustive - contrastive	-0.1876	0.0531	1258	-3.530	.0013 **
	unmodified - contrastive	-0.1402	0.0529	1256	-2.650	.0222 *
total_fixation	exhaustive - unmodified	-0.0265	0.0321	1213	-0.826	.6871
	exhaustive - contrastive	-0.1575	0.0454	1258	-3.471	.0016 **
	unmodified - contrastive	-0.1310	0.0452	1256	-2.898	.0107 *
choice B						
total_dwell	exhaustive - unmodified	0.1579	0.1830	1239	0.862	.6643
	exhaustive - contrastive	0.2904	0.1310	1259	2.215	.0691 .
	unmodified - contrastive	0.1325	0.1420	1249	0.931	.6208
total_fixation	exhaustive - unmodified	0.0497	0.1560	1239	0.318	.9459
	exhaustive - contrastive	0.1289	0.1120	1260	1.151	.4826
	unmodified - contrastive	0.0792	0.1220	1249	0.652	.7914

(and where in processing) contrastive/non-exhaustive readings emerge. Using a sentence–picture verification task with webcam-based eye tracking, we compared unmodified PVF sentences to two explicit baselines: an exhaustive condition marked by *csak* ‘only’ and a non-exhaustive/inclusive condition marked by *többek között* ‘amongst others’.

4.1. Default interpretation of PVF: strong exhaustive bias, but defeasible

The behavioural pattern is clear: unmodified PVF sentences were matched with the exhaustive (single-object) pictures at near-ceiling rates, essentially indistinguishable from the overtly exhaustive *csak* condition (Table 1, Fig. 6). This replicates the robust *exhaustive bias* for PVF reported in earlier experimental work (Gerőcs et al. 2014; Káldi et al.

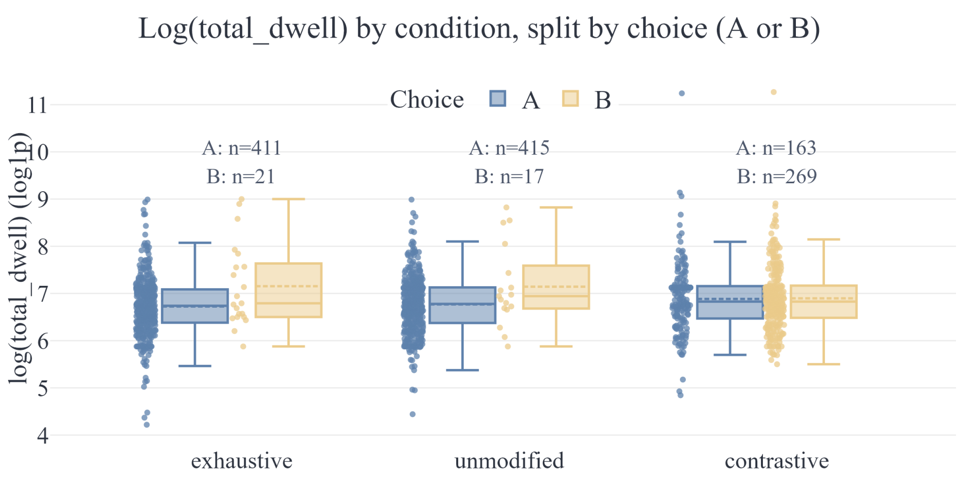

Figure 8. *Total dwell time (log-transformed) by condition and choice type*

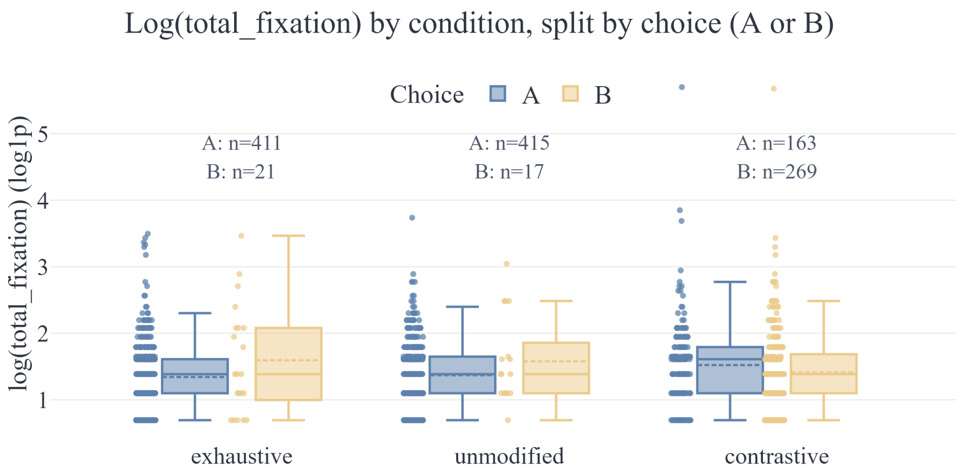

Figure 9. *Total fixation count (log-transformed) by condition and choice type*

2021; Onea 2009), and is consistent with the long-standing intuition in the generative literature that the preverbal focus position is tightly linked to exhaustivity (É.Kiss 1998, 2007; Horvath 2008; Kenesei 2006; Szabolcsi 1981).

At the same time, the contrastive/inclusive manipulation shows that the exhaustive bias is *not* absolute: *többek között* shifted choices toward the non-exhaustive (multi-object) picture, but did not eliminate exhaustive choices (37.7% A-responses in the contrastive condition). This is crucial because it demonstrates that participants can accommodate PVF-like forms with non-exhaustive situations when inclusivity is explicitly licensed, aligning with approaches that locate exhaustivity outside truth-conditional focus semantics (e.g. as pragmatic inference or context-dependent enrichment) (Balogh 2013; Onea & Beaver 2009; Wedgwood 2006; Zimmermann 2008). In other words, the present data support a *default-exhaustive but contextually defeasible* characterization of PVF: the unmodified construction strongly *invites* exhaustification in neutral contexts, but can be overridden when discourse cues make alternatives salient or explicitly include additional true alternatives.

This pattern is broadly compatible with alternative-based theories of focus (Krifka 2008; Rooth 1992; Vallduví & Vilkuna 1998). PVF can be taken to reliably activate a salient set of alternatives; in a minimal context like the current task, participants appear to resolve this competition by selecting the strongest construal (exhaustification), yielding an ‘‘only-like’’ outcome even without an overt exclusive. When an inclusive cue (*többek között*) is present, the same alternative set is still available, but the discourse instruction shifts: additional true alternatives must be compatible with the event description, making the non-exhaustive picture a better match overall.

4.2. Why do exhaustive choices persist under többek között?

The residual rate of exhaustive choices in the contrastive condition is informative. One possibility is that participants sometimes treated *többek között* as compatible with a *covert* larger set that is not depicted (e.g. ‘‘among other things (in general)’’), thereby allowing a single-object scene to be accepted as a partial depiction rather than a complete model of the described situation. Another possibility is task-driven: the forced choice between two pictures may encourage a ‘‘best match’’ strategy even when neither picture perfectly instantiates all aspects of the sentence meaning. Under this interpretation, some participants may still prefer the single-object image because it matches the focused object most transparently (especially when PVF already promotes an exhaustive construal), while treating the inclusive phrase as secondary or as introducing weak, backgrounded information. Both explanations are compatible with the broader point that exhaustivity in PVF behaves like a strong default rather than an entailment.

4.3. Processing locus: verification conflict rather than incremental sentence decoding

Self-paced reading did not show systematic condition differences across sentence regions, suggesting that participants did not experience reliable difficulty differences during incremental decoding of the written sentence. In contrast, the eye-tracking measures during picture verification *did* differentiate conditions, and crucially did so in a way that depends on the interpretation participants ultimately committed to.

For total dwell time and total fixation count, the mixed-model results show a reliable Condition×Choice interaction (Table 12). The most consistent contrast is that *when participants chose the exhaustive (A-type) image*, gaze time and fixation counts were higher in the contrastive condition than in the exhaustive and unmodified conditions (Table 14). This pattern is naturally interpreted as a *verification conflict*: selecting an exhaustive scene after reading an explicitly inclusive sentence requires either (i) reanalysis of the linguistic cue, or (ii) accommodation of a broader ‘‘amongst others’’ set not represented visually, both of which plausibly increase decision time and re-inspection of the images. Conversely, when participants chose the multi-object (B-type) image, condition differences were weak or absent, suggesting that the contrastive image is relatively easy to verify once the non-exhaustive construal is adopted.

Taken together, the absence of robust reading-time effects and the presence of verification-stage gaze effects point to a division of labour: the PVF/*only*/*amongst others* distinctions do not strongly disrupt basic sentence comprehension in this paradigm, but they do modulate the competition between candidate situation models during *sentence–scene mapping*. This is in line with the idea that exhaustivity-related inferences may be computed (or strengthened) when the comprehender must decide among explicit alternatives (cf. Gerőcs et al. 2014; Onea 2009).

4.4. Implications for the exhaustivity debate

The present findings do not straightforwardly support a view on which PVF *semantically entails* exhaustivity in the same way as an overt exclusive operator, because the experiment demonstrates interpretive flexibility once inclusivity is explicitly cued. At the same time, the data also do not reduce PVF exhaustivity to a weak, easily cancellable conversational inference: the unmodified PVF condition patterns extremely closely with *csak* in behavioural choice, and the ‘‘exhaustive pull’’ remains visible even in the contrastive condition. This combination is best captured by accounts that separate (i) *alternative activation* and strong expectations associated with the Hungarian preverbal focus position (É.Kiss 1998; Gécseg 2020; Krifka 2008) from (ii) *operator-level exhaustification* contributed by *csak* (Balogh 2007), while allowing discourse and task structure to determine how strongly exhaustification is endorsed in a given decision.

4.5. Limitations and future directions

Several limitations constrain the generalization of these results. First, the paradigm is explicitly metalinguistic in the sense that it requires selecting between two designed interpretations; this may amplify exhaustification compared to natural conversation, where the space of alternatives is not made visually explicit. Second, the stimuli were presented in writing, while Hungarian PVF is tightly intertwined with prosody (e.g. eradicating stress and post-focus deaccentuation) (Gécseg 2020; Genzel et al. 2015; Kornai & Kálmán 1988). Future work should therefore test the same manipulation with spoken stimuli and controlled intonation to assess whether prosody strengthens, weakens, or reshapes the observed default. Third, the webcam-based eye-tracking signal is necessarily coarse, and the current AOI design collapses gaze to left/right canvases; richer spatial AOIs and higher-resolution tracking could reveal more fine-grained time courses (e.g. early preference vs late reconsideration). Finally, discourse context was minimal by design; extending the study with explicit Questions Under Discussion, corrective dialogues, or contrast sets would help determine when PVF yields genuinely contrastive-but-non-exhaustive interpretations without an overt inclusive licenser.

4.6. Interim conclusion

Overall, the converging behavioural and eye-tracking evidence supports a robust default-exhaustive interpretation for unmodified Hungarian PVF in neutral contexts, while confirming that non-exhaustive construals are available and become behaviourally prominent when inclusivity is explicitly licensed. Processing-wise, the main cost appears during sentence–scene verification, especially when participants commit to an exhaustive interpretation despite an inclusive cue, consistent with a competition/accommodation account of how exhaustivity and contrastivity interact in PVF.

…

Tie all this back to the literature!!!

5. Conclusion

6. NOTES AND MISC

Table 15. *Model-estimated total dwell time during choice (ms; back-transformed from* $log(1+x)$ *EMMs). Values are estimated marginal means and (standard errors).*

Condition	Choice A: EMM (SE)	Choice B: EMM (SE)
Exhaustive	836 (54)	1299 (181)
Unmodified	870 (56)	1107 (166)
Contrastive	1002 (75)	972 (67)

Table 16. *Model-estimated total fixation count during choice (back-transformed from* $log(1+x)$ *EMMs). Values are estimated marginal means and (standard errors).*

Condition	Choice A: EMM (SE)	Choice B: EMM (SE)
Exhaustive	2.86 (0.20)	3.71 (0.55)
Unmodified	2.94 (0.20)	3.48 (0.57)
Contrastive	3.48 (0.27)	3.14 (0.23)

Table 15 reports model-estimated total dwell time during the choice phase, separated by sentence condition and by the type of image participants selected (A = exhaustive/single-object; B = contrastive/multi-object). In the Exhaustive and PVF conditions, dwell times were lower for exhaustive choices (A) than for contrastive choices (B), whereas in the Contrastive condition dwell times were comparable across the two choice types. Overall, the pattern indicates that verification time depends on the combination of focus condition and the interpretation participants commit to at response.

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Data availability statement. Replication data and code can be found at https://doi.org/link.

Appendix

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Figure 10. *Complete visual stimulus set with A) single- and B) multi-entity image pairs*